3 A boy fills a flask with fluid and places it on a mass balance. The boy records the reading on the mass balance x in grams. He inserts two similar cubes, one on top of the other, into the flask as shown in Fig. 3.1.

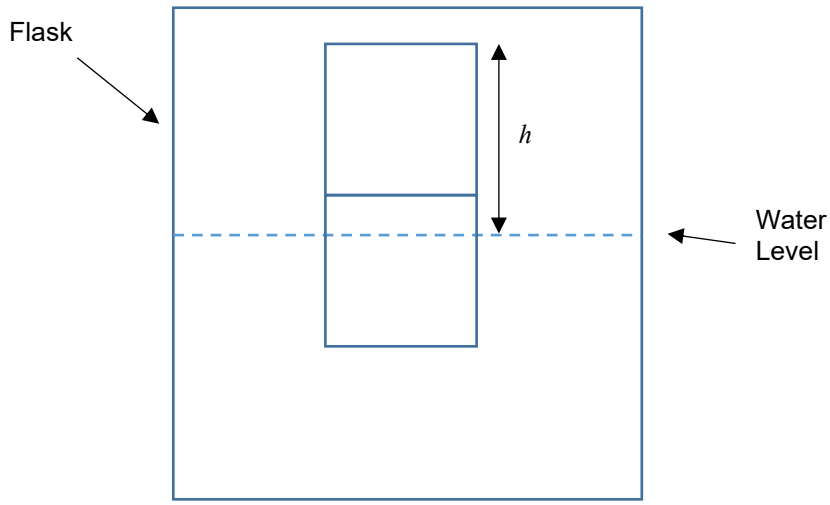

Fig. 3.1

Each side of the cube is r in cm. Both cubes stay afloat with one cube being partially immersed. After some time, the boy records that height of the two cubes above the fluid surface is h in cm while the reading on the mass balance is now y in grams.

(a) Express the density of the fluid in terms of x, y, r and h in g cm^{-3}.

[2]

(b) The cube on top is removed. Suggest and explain the subsequent motion of the cube in the fluid.

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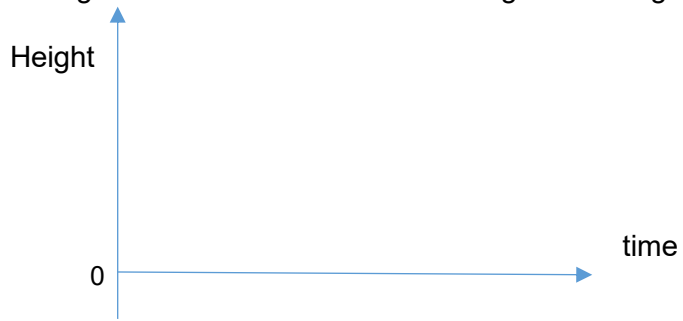
.....[4]

[Turn over

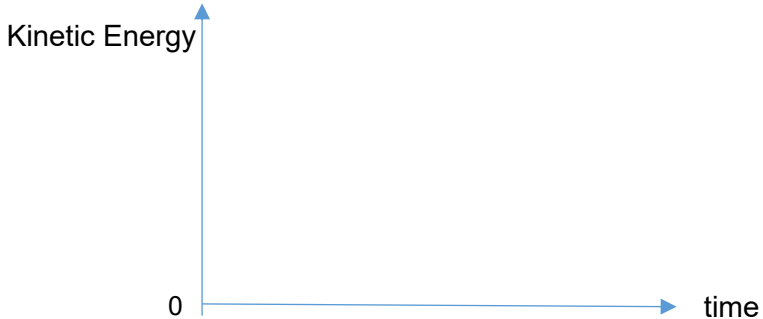
3(c) Given the values of r = 2.00 cm and h = 3.30 cm, sketch, without numerical values, the following graphs for the cube that remains in the fluid. Note that the bottom cube has a height of 1.30 cm above the fluid surface before the top cube is removed.

(i) Height of cube above fluid surface against time graph,



[2]

(ii) Kinetic Energy against time graph.



[2]

4 (a) The *I-V* characteristics of resistor **R** and resistor **X** are investigated. The results